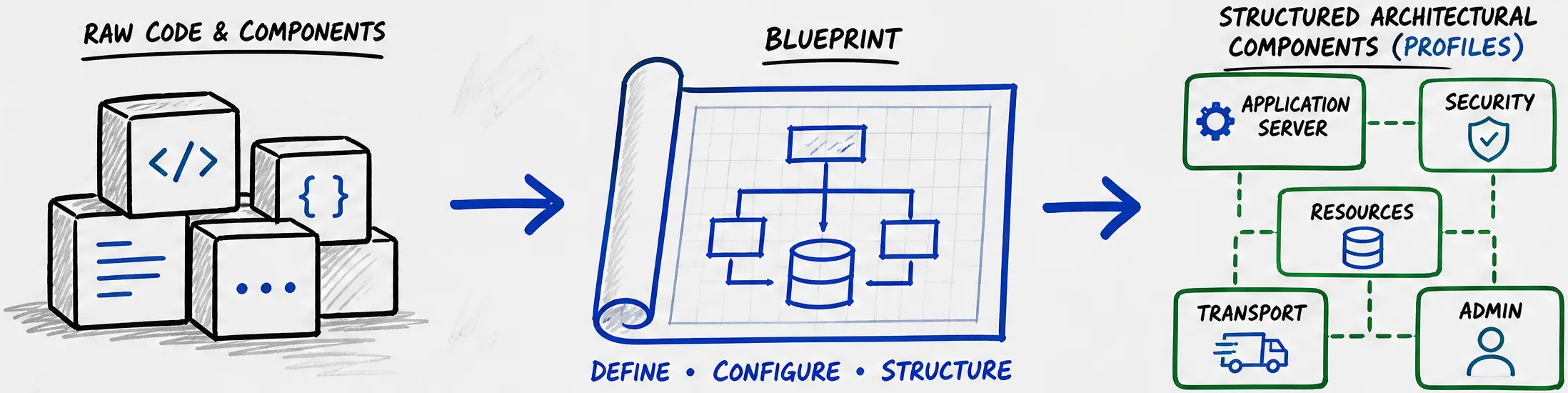
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WebSphere v9 Administration: Module 2

Profiles Build the Runtime



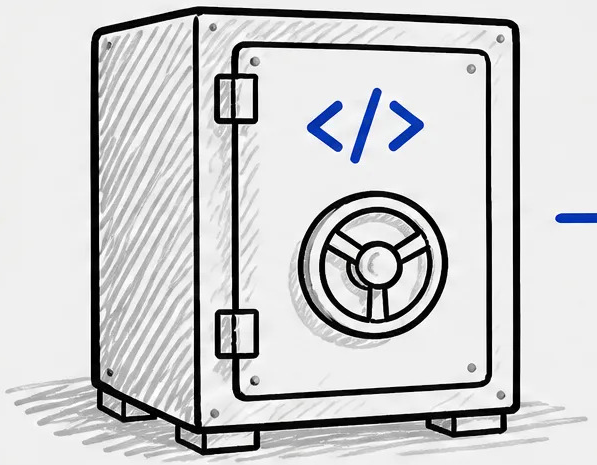
Profiles turn installed software into an operating environment.

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Core Files vs. Profiles

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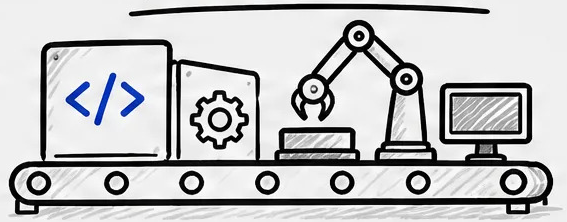
CORE FILES
(CORE BINARIES)



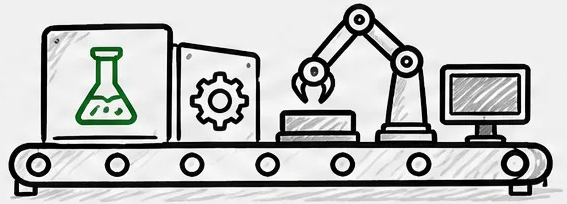
SINGLE SOURCE OF TRUTH

INSTALL ONCE, CONFIGURE MANY

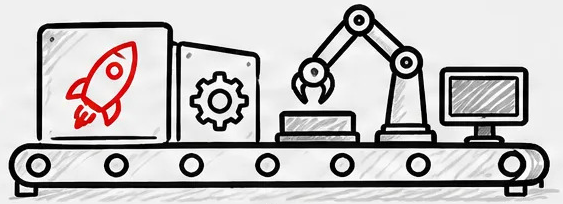
PROFILES
(DISTINCT RUNTIMES)



DEVELOPMENT
- Config A
- Settings
- Libraries



QA TESTING
- Config B
- Settings
- Libraries



PRODUCTION
- Config C
- Settings
- Libraries

ONE PRODUCT HOME CAN SUPPORT MANY DELIBERATELY DISTINCT RUNTIMES.

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The Profile Family

Six Paths to Topologies

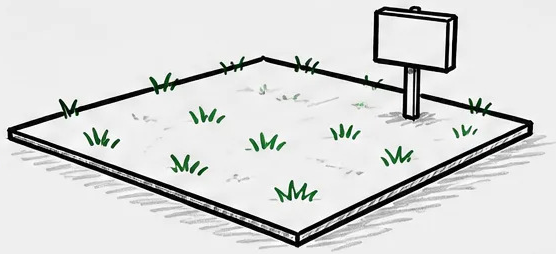
1 App Server



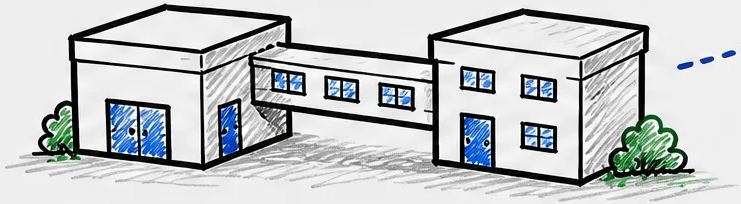
2 DMGR



3 Custom



4 Cell



5 Admin Agent



6 Job Manager

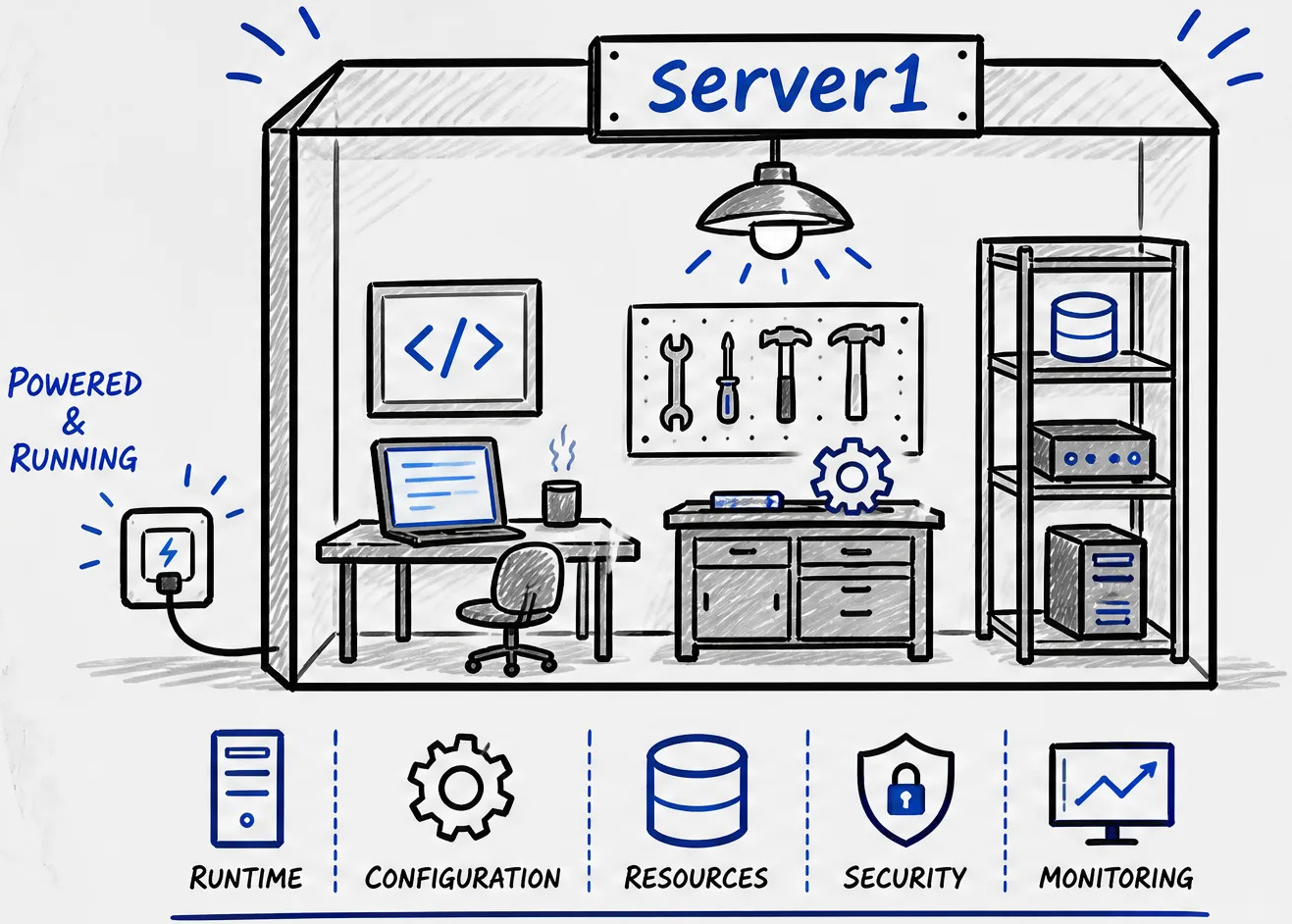


Choose the profile by the **topology you need**, **not** by the screen you happen to see.

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Application Server Profile



SELF-CONTAINED EXECUTION

A stand-alone server is independent by design, not unmanaged by accident.

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Deployment Manager Profile

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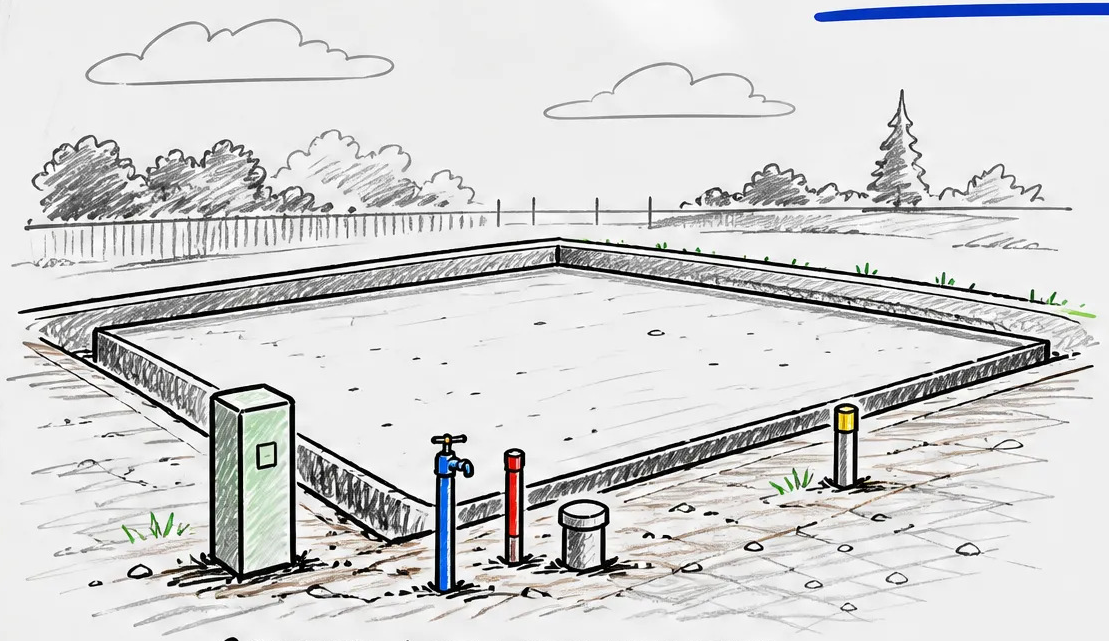


Central administration is **valuable only when** the topology is **intentionally federated.**

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Custom Profile



GRADED LOT WITH UTILITIES READY, STRUCTURE TO COME.

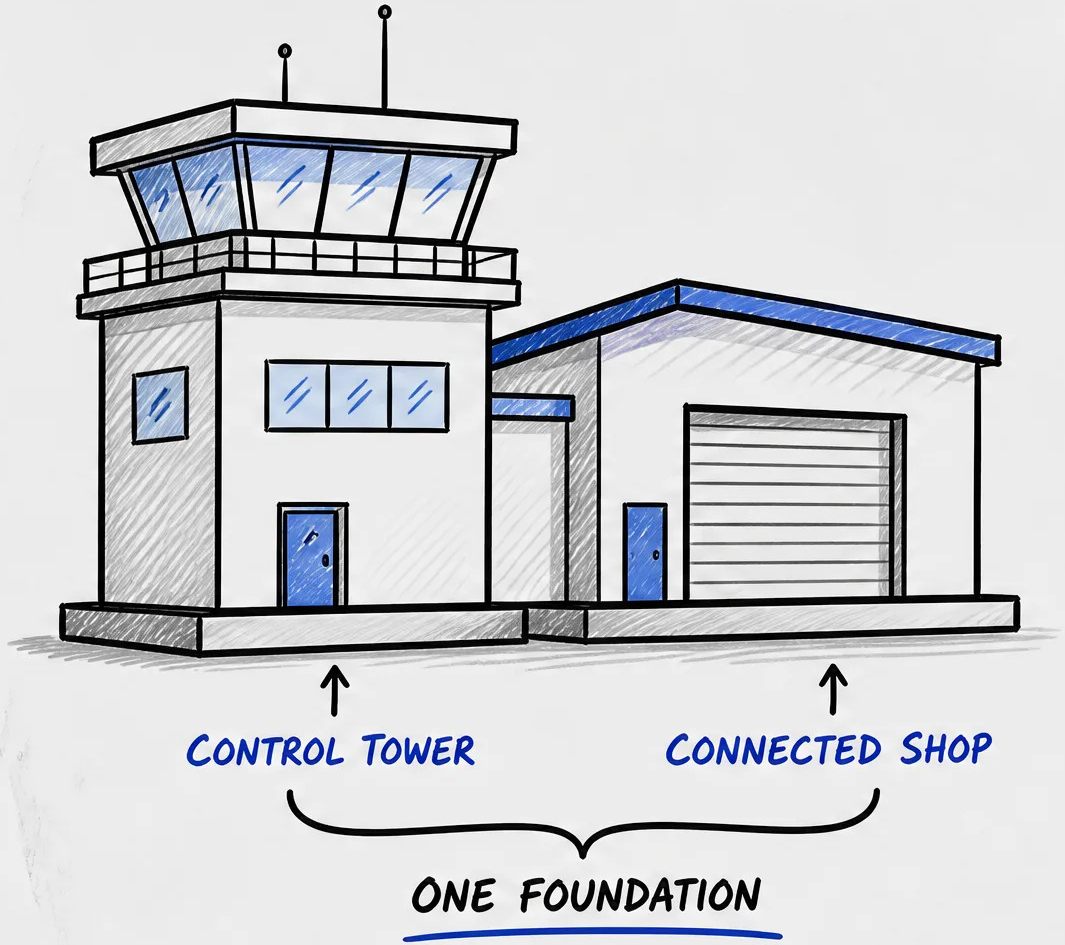
PREPARED CAPACITY

CUSTOM PROFILES ARE **CAPACITY** WITHOUT A **PREBUILT WORKLOAD.**

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Cell Profile

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Instant Convenience

Convenience is not a production topology.

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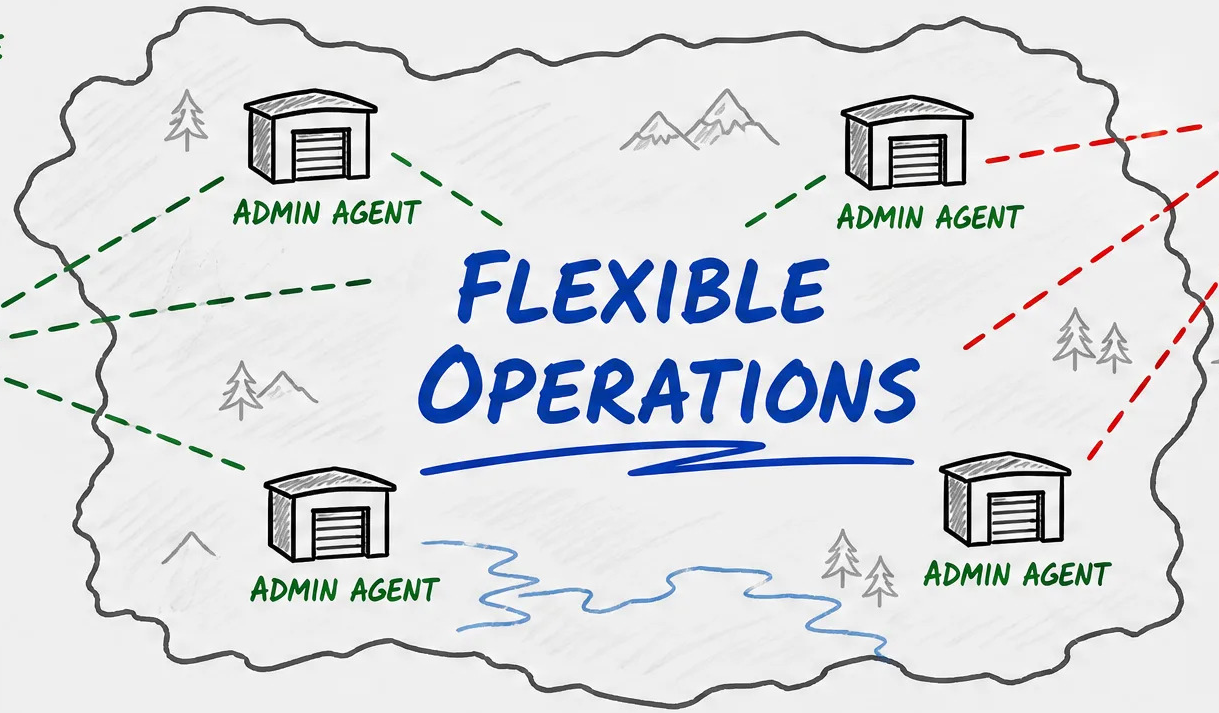
Administrative & Job Manager Profiles

Administrative Profile (Admin Agent)



Local Foreman

Oversees multiple independent shops across the region



Job Manager Profile (Job Manager)



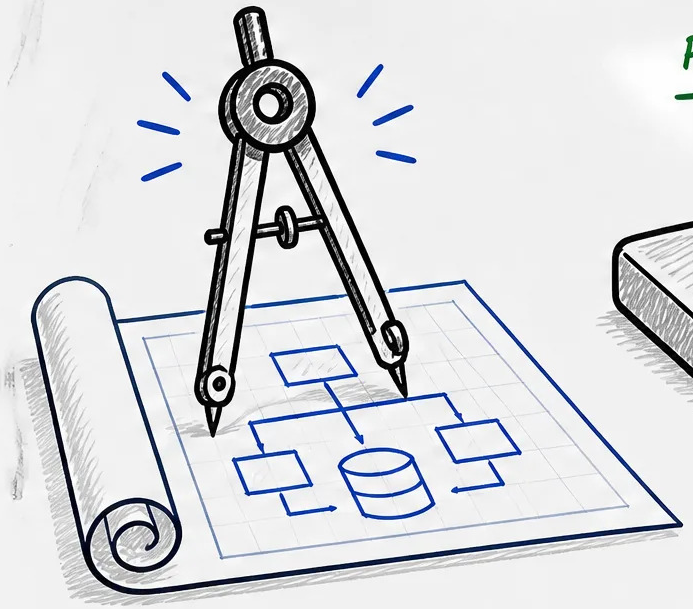
Radio Tower

Pings updates across distances to all shops



Shared visibility is NOT the same as a shared cell.

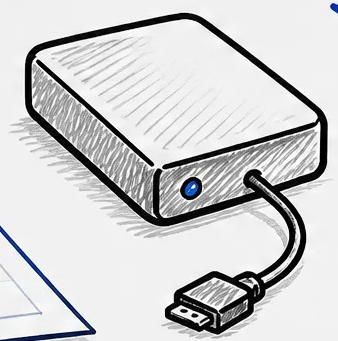
Planning the Profile Estate



Architecture Design

- Define structure
- Map relationships
- Plan for growth

Space Planning



- Capacity planning
- Resource allocation
- Performance headroom

Design Before You Build

Security by Design



- Access control
- Isolation & trust
- Compliance ready

Nametag Strategy



- Naming standards
- Consistent tagging
- Clear ownership



Planning is the first configuration setting.

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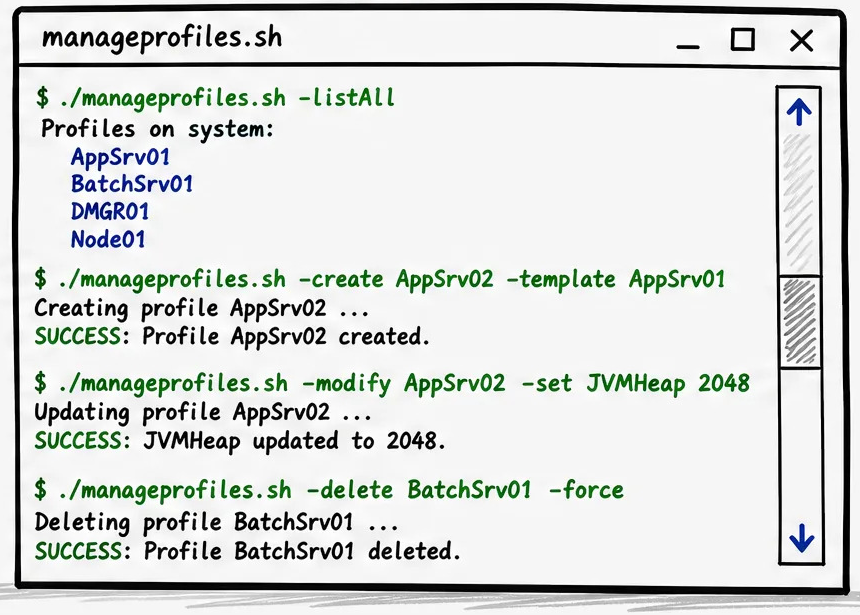
PMT and manageprofiles

PMT



Visualizing the Commands

manageprofiles



The GUI teaches the choices; the command line scales the choices.

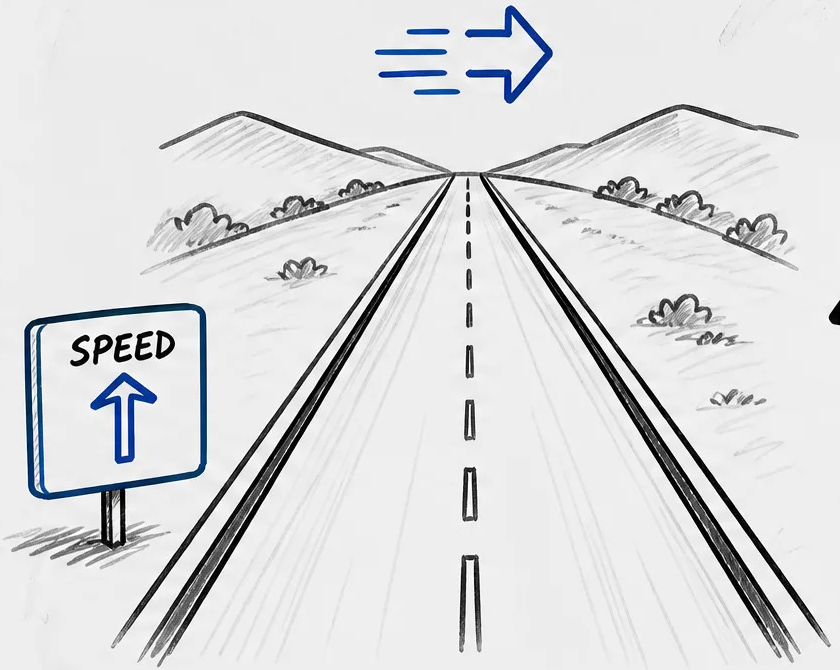
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Typical vs. Advanced Creation

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Typical

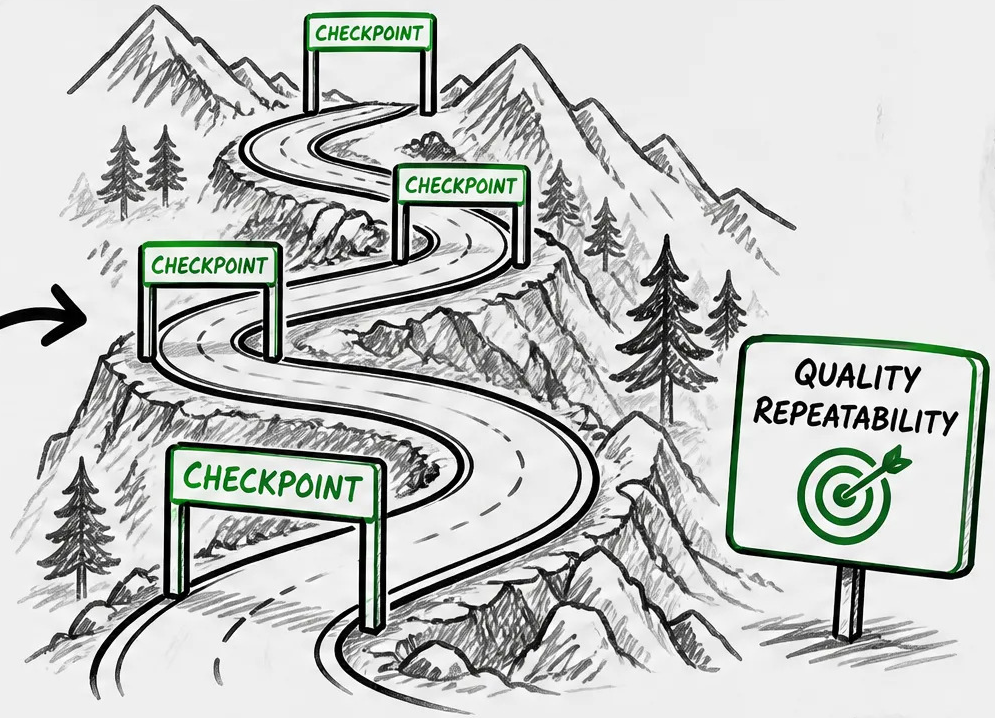
Fast & Straight



Control over Speed

Advanced

Controlled & Repeatable

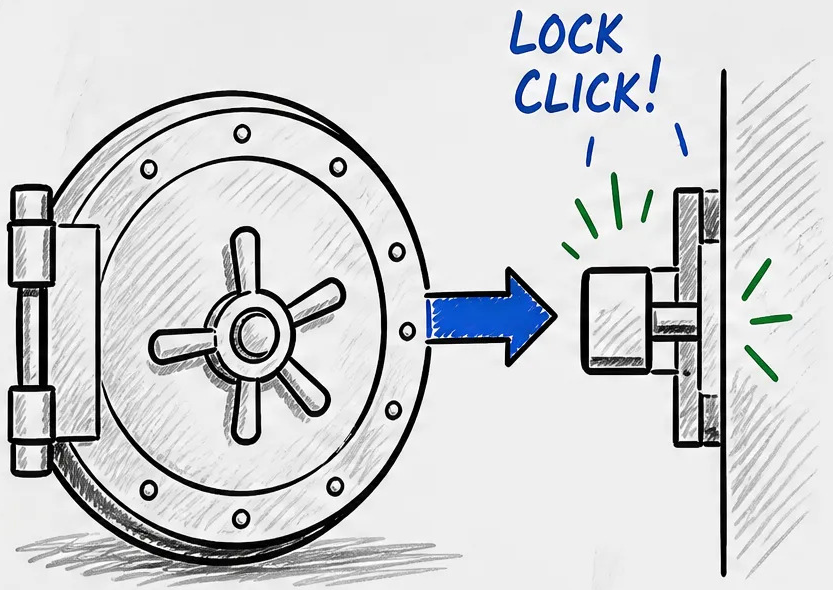


Repeatability beats both speed and preference.

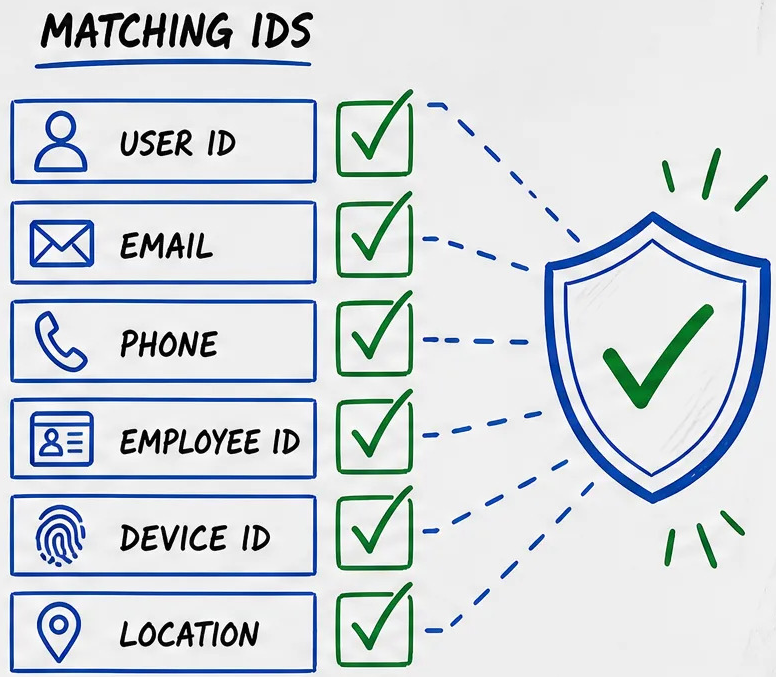
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Security at Profile Creation

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Inconsistent **Security** is an Architecture Defect



Inconsistent **Security** is an **Architecture** **Defect**, not a **Login Problem**.

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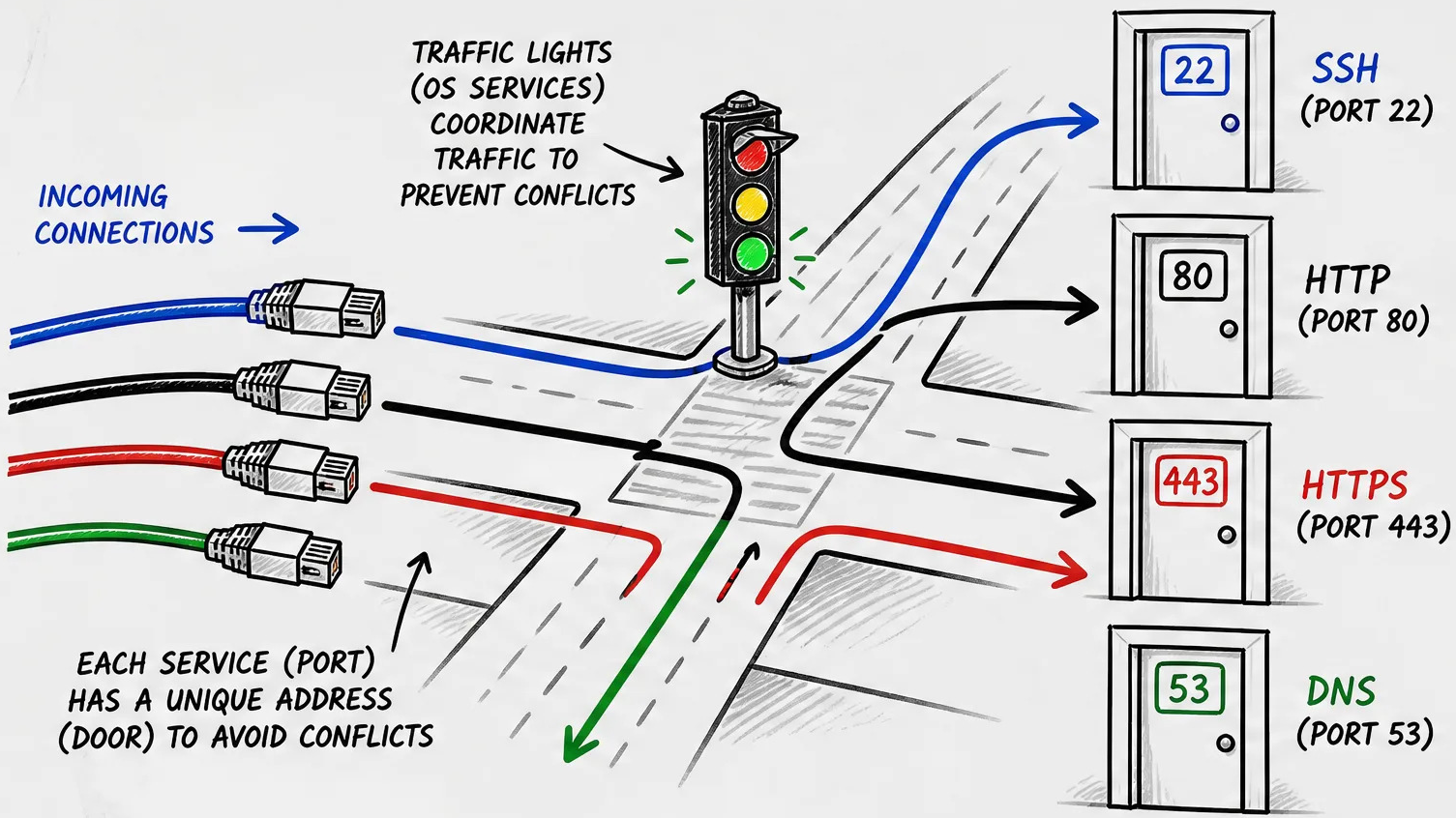
Certificates and Trust



Memorable Statement:
Connectivity without trust is not a **management channel.**

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Ports and OS Services



Port Conflicts Are Topology Errors

Port conflicts are topology errors expressed as startup failures.

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Verify Before You Build

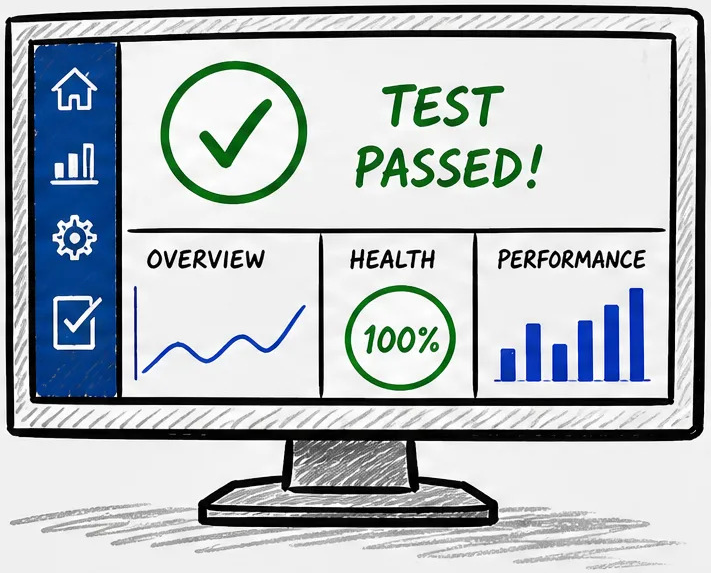
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VERIFY THE RUNBOOK



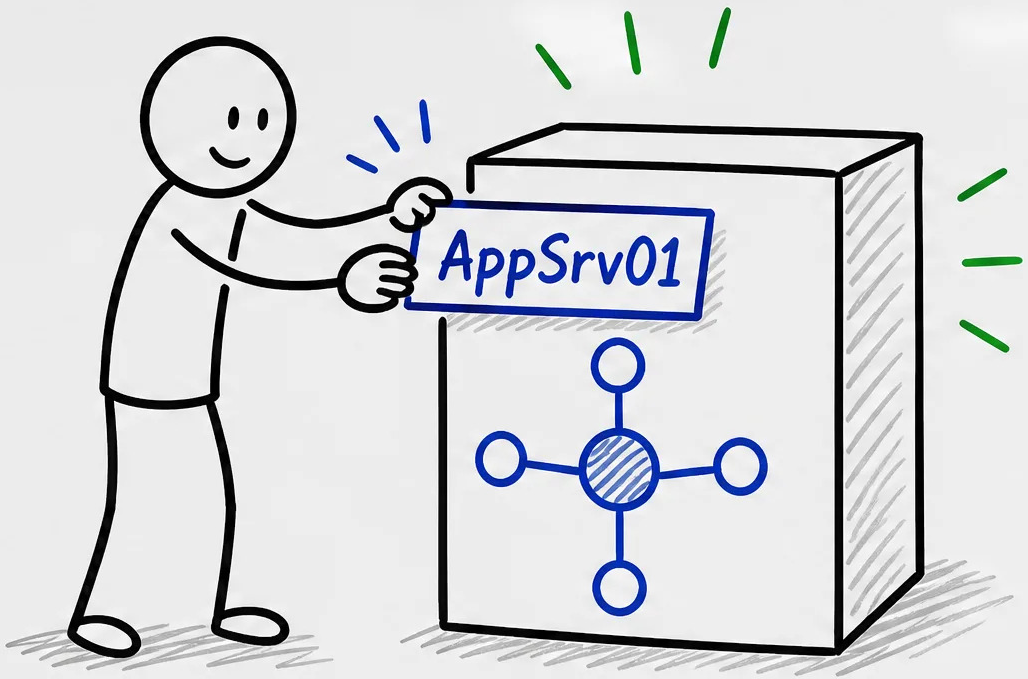
VERIFICATION IS THE END OF THE RUNBOOK

SUCCESSFUL TEST RUN



A PROFILE IS COMPLETE ONLY WHEN THE INTENDED CONTROL AND DATA PATHS WORK.

Creating an App Server Profile



Default is Just a Routing Behavior

Default is a **Routing Behavior**, not an **ownership model**.

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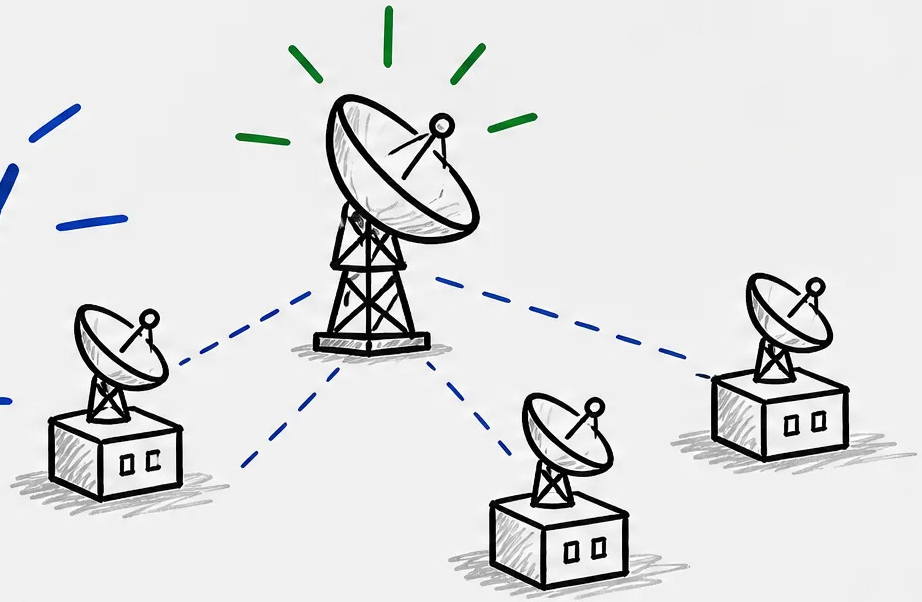
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Creating a Deployment Manager



ERECTING THE PRIMARY DEPLOYMENT MANAGER

TOPOLOGY FIRST



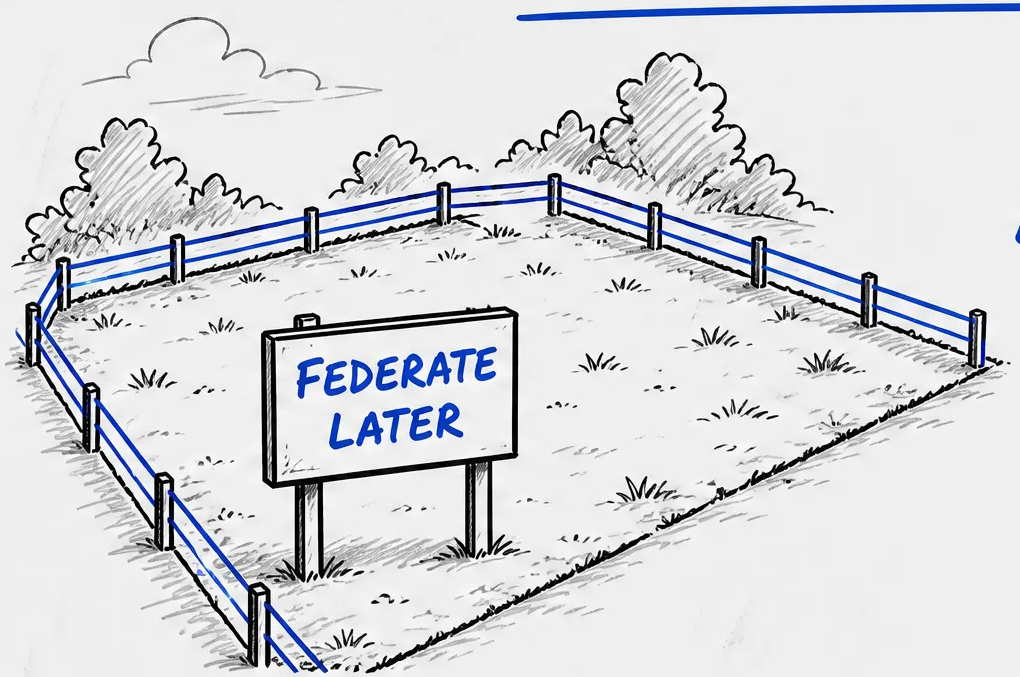
CONNECTING TO EXISTING NODES

A CELL BEGINS WITH A REACHABLE SOURCE OF ADMINISTRATIVE TRUTH.

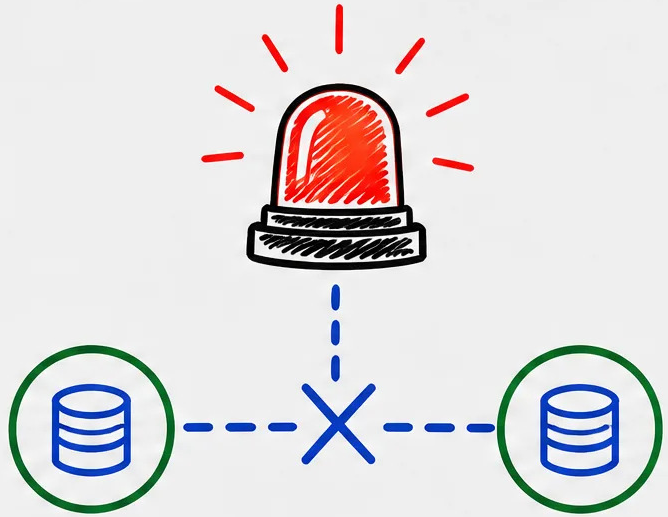
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Creating a Custom Profile



A Clean Delay Is Safer





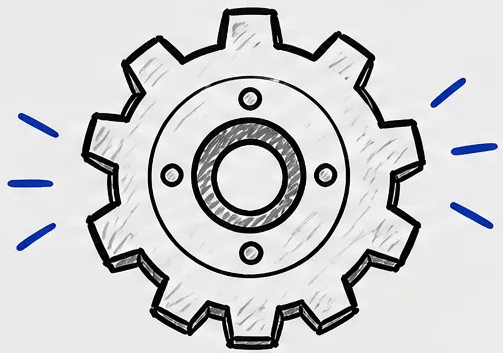
A Clean Delayed Federation is -SAFER than a BROKEN IMMEDIATE federation.

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Federation with addNode

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NODE

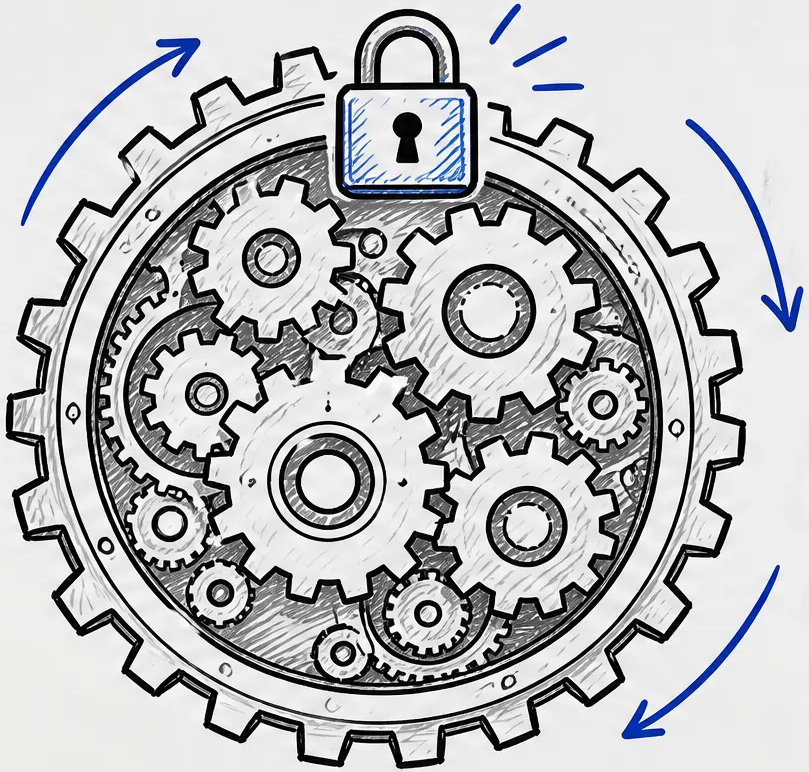


Smaller entity seeking to join the cell

Transfer of Authority

Federation is a transfer of administrative authority, not merely a network connection.

CELL



Larger administrative domain (cell) with established governance

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THE WIZARD — IS JUST THE — BEGINNING

THE WIZARD IS ONLY THE MIDDLE OF THE RUNBOOK;
VERIFICATION IS THE END.

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Admimistrative-Agent Management

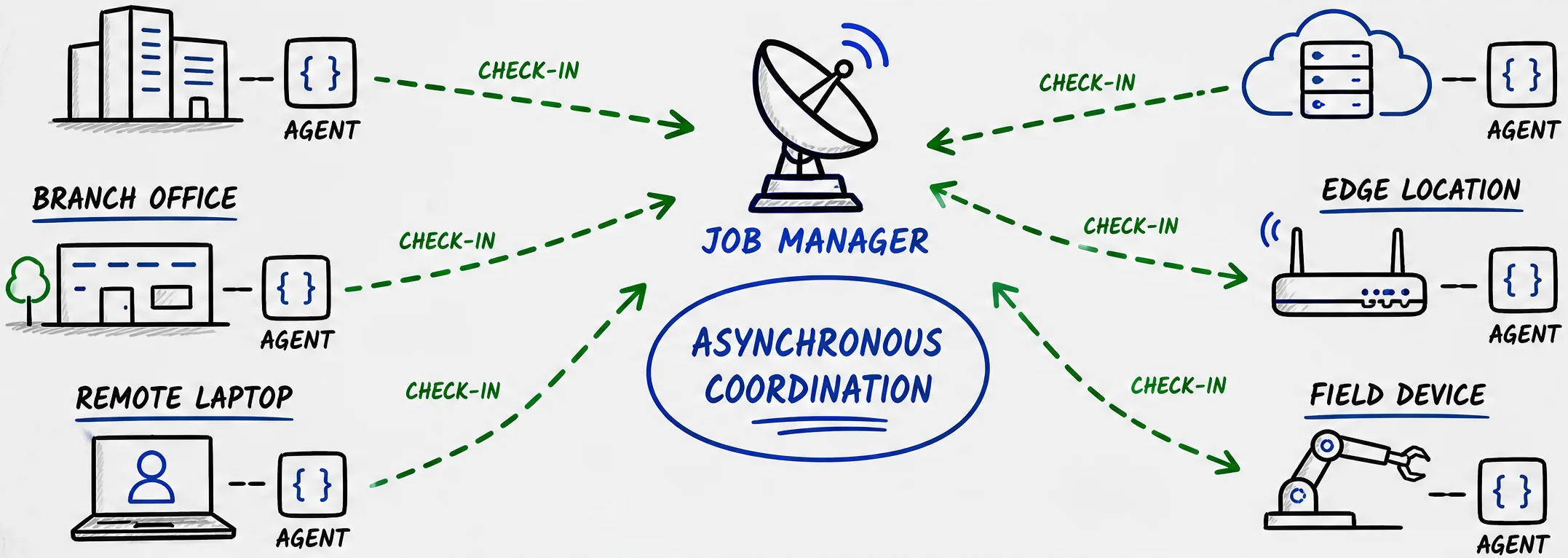


FLEXIBILITY WORKS ONLY WHEN THE MANAGEMENT CAPABILITY MATCHES THE MANAGED CAPABILITY.

Registering with a Job Manager

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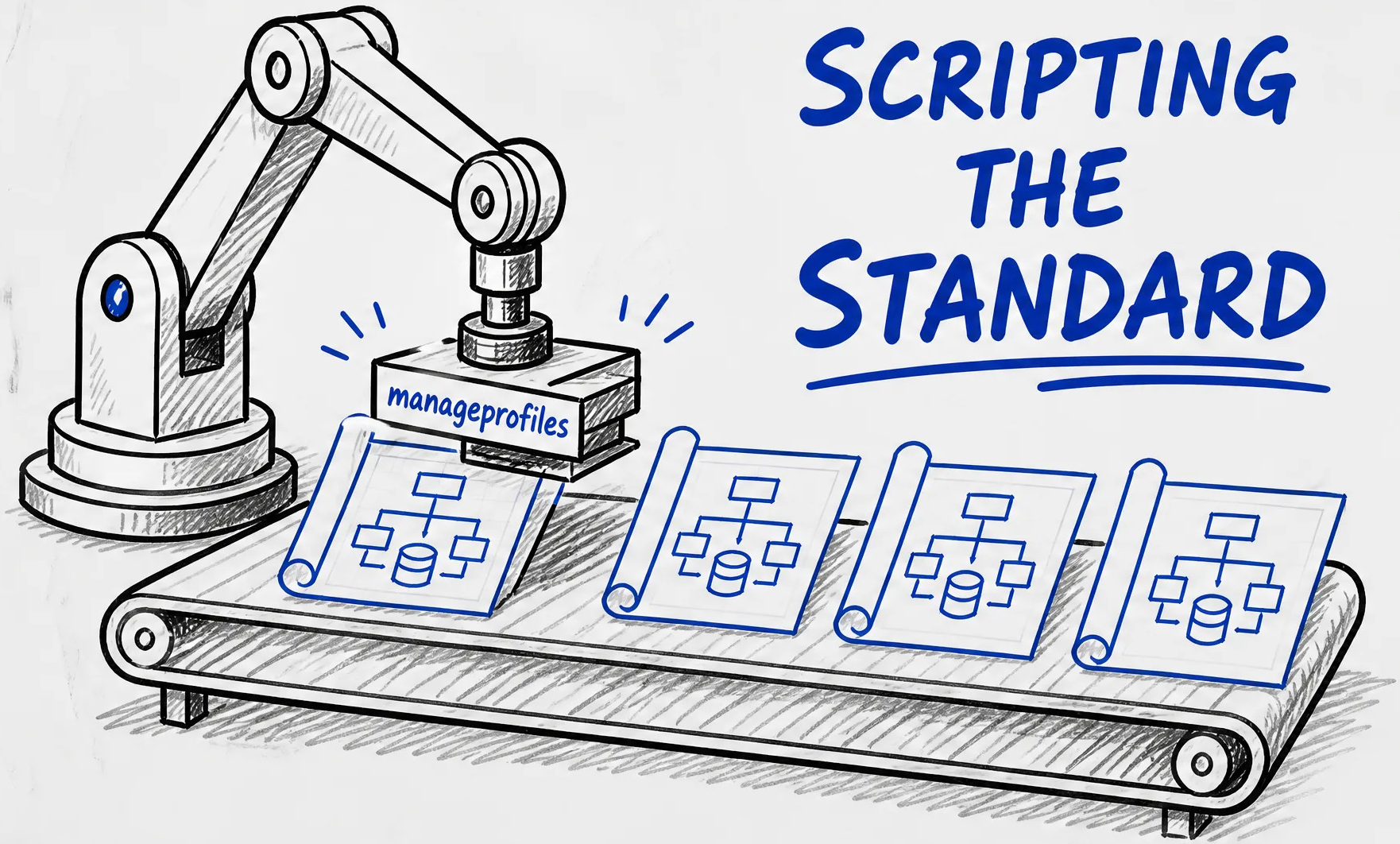
DISPARATE LOCATIONS. ONE JOB MANAGER.
REGISTER, CHECK IN, STAY COORDINATED.



REGISTRATION CREATES A MANAGED RELATIONSHIP;
IT DOES NOT MERGE THE ENVIRONMENTS.

Automation and Templates

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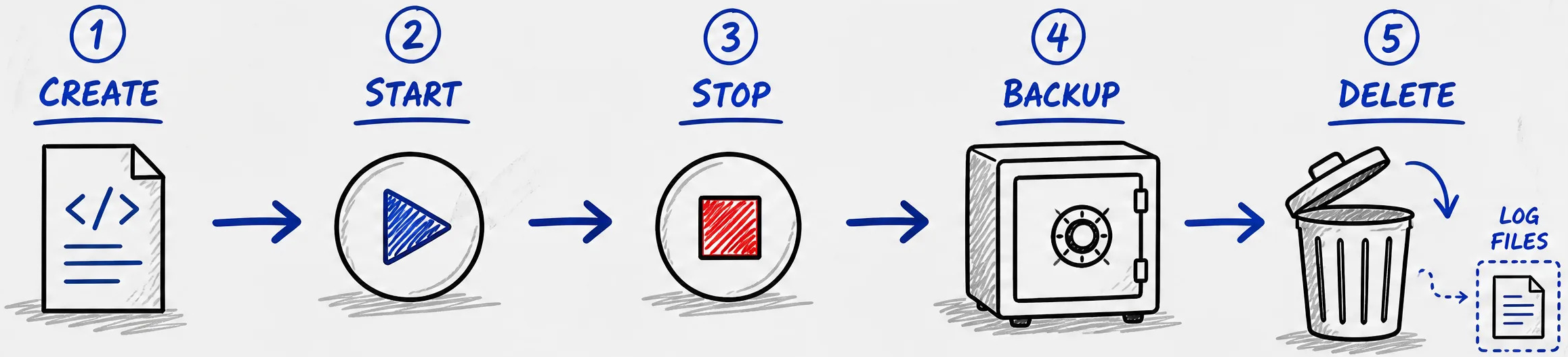


SCRIPTING THE STANDARD

MEMORABLE STATEMENT

Automation converts a **profile recipe** into a **repeatable asset**.

Lifecycle: Backup and Delete



End-to-End Responsibility

The profile lifecycle is the foundation of every reliable WebSphere topology.